HAT-P-32b

R-0141

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-32_231027` · created 2026-07-16T07:36:53+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460244.8266 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1861 +/- 0.0076
Transit depth (Rp/Rs)^2	3.46 +/- 0.28 [%]
Orbital Inclination (inc)	89.8 +/- 1.48
Airmass coefficient 1 (a1)	2.286 +/- 0.074
Airmass coefficient 2 (a2)	-0.715 +/- 0.056
Scatter in the residuals of the lightcurve fit is	3.22 %
Best Comparison Star	None
Optimal Aperture	14.21
Optimal Annulus	30.86
Transit Duration (day)	0.1343 +/- 0.0013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1861 ± 0.0076 vs 0.1489 (deviation 2.46σ)
Duration fit vs published	0.1343 d vs 0.1304 d (deviation 1.5σ)
Mid-time O–C	10.8 min
Depth SNR	12.3
Residual scatter	3.22 %

Light curve

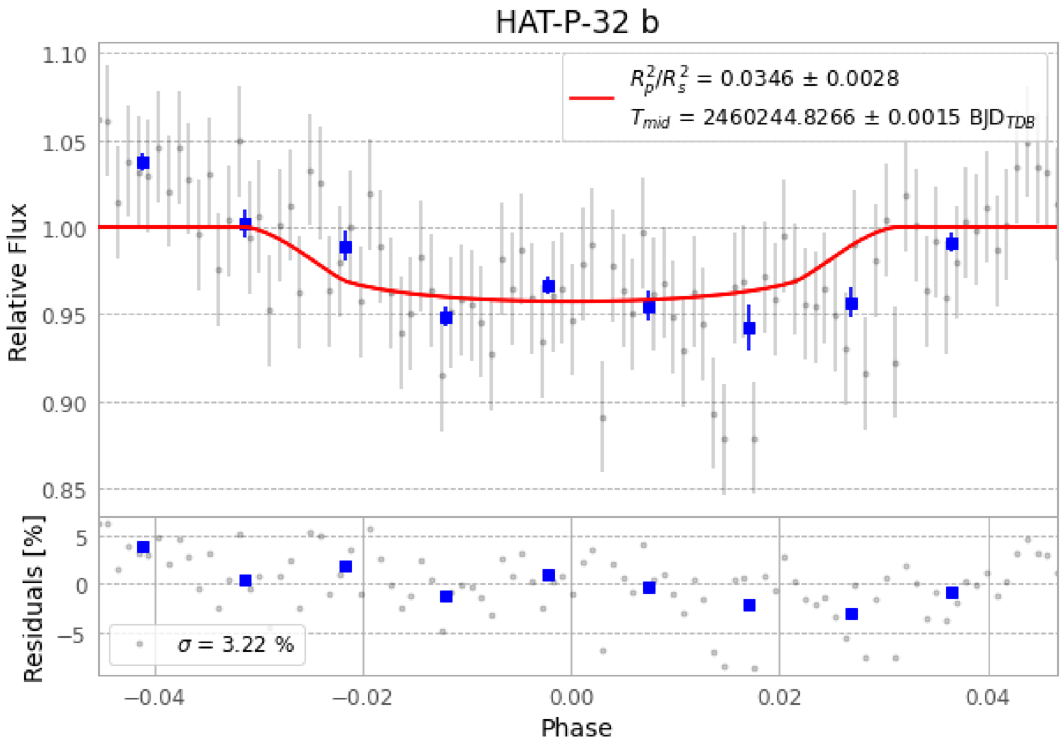

Figure 1 — FinalLightCurve_HAT-P-32 b_2023-10-26.png (SHA-256 88d9e304ccb0520d…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [21, 65], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 82ab821c20fabae8…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba`

Data provenance

96 FITS frames (63 MB), `HATP-32231027052115.FITS` → `HATP-32231027100615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-231027.log (9d84b047521b…), FinalLightCurve_HAT-P-32 b_2023-10-26.png (88d9e304ccb0…), FinalLightCurve_HAT-P-32 b_2023-10-26.csv (dfb40b7069fc…)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)